

DATA ENGINEERING INTERVIEW PREPARATION GUIDE

SQL • Python • Spark • Hadoop • AWS | Freshers / LTM-Style Preparation

How to answer in an interview	Best structure
Definition	Start with one clear sentence.
Difference / purpose	Explain what it does and when it is used.
Example	Give a small practical example if asked.
Follow-up	Be ready for performance, limitations, or scenario questions.

Interview rule: Don't memorize long paragraphs. Remember the key idea, then explain it naturally.

1. SQL Interview Questions

For each question: answer in 1–2 sentences first. Expand only if the interviewer asks a follow-up.

1. What is the difference between INNER JOIN and LEFT JOIN?

Answer: INNER JOIN returns only matching rows from both tables; LEFT JOIN returns every row from the left table plus matching rows from the right.

2. What is a Primary Key?

Answer: A primary key uniquely identifies each row and does not allow NULL values.

3. Difference between WHERE and HAVING?

Answer: WHERE filters rows before grouping/aggregation; HAVING filters groups after aggregation.

4. What is a Subquery?

Answer: A subquery is a query nested inside another SQL statement and can be used for filtering, calculation, or comparison.

5. What is a CTE?

Answer: A Common Table Expression is a named temporary result set defined using WITH and used within a single SQL statement.

6. How does GROUP BY work?

Answer: GROUP BY groups rows having the same values so aggregate functions such as COUNT, SUM, AVG, MIN, and MAX can be applied.

7. What is a Window Function?

Answer: It calculates across related rows without collapsing the original rows, using an OVER clause.

8. RANK() vs DENSE_RANK()?

Answer: RANK skips numbers after ties; DENSE_RANK gives consecutive ranks even when values are tied.

9. What is normalization?

Answer: Normalization organizes data into related tables to reduce redundancy and improve data integrity.

10. What is denormalization?

Answer: Denormalization intentionally adds redundancy or combines data to improve read/query performance.

11. What is a foreign key?

Answer: A foreign key references a key in another table and helps maintain relationships and referential integrity.

12. How do you find duplicates?

Answer: GROUP BY the suspected duplicate columns and use HAVING COUNT(*) > 1.

13. Use of LIMIT or TOP?

Answer: They restrict the number of rows returned; LIMIT is common in MySQL/PostgreSQL and TOP is common in SQL Server.

14. How do you calculate a running total?

Answer: Use SUM(column) OVER (ORDER BY date_column) to calculate a cumulative total.

15. What is NULL?

Answer: NULL represents missing, unknown, or unavailable data; it is not the same as zero or an empty string.

16. What does COALESCE() do?

Answer: COALESCE returns the first non-NULL expression from its arguments.

17. UNION vs UNION ALL?

Answer: UNION combines compatible result sets and removes duplicates; UNION ALL keeps duplicates and is generally faster.

18. What are indexes?

Answer: Indexes are data structures that speed up reads but add storage and can increase insert/update/delete overhead.

19. DELETE vs TRUNCATE?

Answer: DELETE removes selected rows and is transactional in many databases; TRUNCATE removes all rows efficiently, with rollback behavior depending on the database.

20. Default ORDER BY direction?

Answer: Ascending (ASC) is the default sort direction.

21. How do you convert date formats?

Answer: Use CAST, CONVERT, DATE_FORMAT, TO_DATE, or other database-specific date functions.

22. IN vs EXISTS?

Answer: IN compares against a set of values; EXISTS checks whether the subquery returns at least one row.

23. What is a view?

Answer: A view is a stored SQL query exposed as a virtual table; a normal view does not store its own result data.

24. Can we index a view?

Answer: Some databases support indexed or materialized views with database-specific restrictions; a normal view itself is not an indexable storage object.

25. What are ACID properties?

Answer: Atomicity, Consistency, Isolation, and Durability ensure reliable transaction processing.

2. Python Interview Questions

For each question: answer in 1–2 sentences first. Expand only if the interviewer asks a follow-up.

1. List vs Tuple?

Answer: Lists are mutable; tuples are immutable. Both can contain ordered collections of objects.

2. What is a dictionary?

Answer: A dictionary stores key-value pairs and is commonly written using curly braces, for example `{'id': 101}`.

3. How is memory managed in Python?

Answer: Python uses automatic memory management, including reference counting and garbage collection for cyclic references.

4. Use of *args and **kwargs?

Answer: *args accepts variable positional arguments; **kwargs accepts variable keyword arguments.

5. What is a lambda function?

Answer: A lambda is a small anonymous function written with the lambda keyword.

6. is vs ==?

Answer: is checks object identity; == checks whether values are equal.

7. Common Python data types in data engineering?

Answer: Common types include int, float, str, bool, list, tuple, dict, set, and datetime objects.

8. How to read a CSV?

Answer: Using pandas, commonly with `pd.read_csv('file.csv')`.

9. How to handle missing values in pandas?

Answer: Use `isna/isnull` to detect them and `dropna` or `fillna` to remove or replace them.

10. What is a generator?

Answer: A generator produces values lazily using `yield`, which can reduce memory usage for large datasets.

11. What is list comprehension?

Answer: A compact syntax for creating a list from an iterable, optionally with a condition.

12. map() vs filter()?

Answer: `map` transforms each item; `filter` keeps items that satisfy a condition.

13. What are decorators?

Answer: Decorators wrap functions or classes to extend or modify their behavior without changing the original code directly.

14. copy vs deepcopy?

Answer: `copy` creates a shallow copy; `deepcopy` recursively copies nested mutable objects.

15. How do you handle exceptions?

Answer: Use `try/except`, optionally `else` and `finally`, and `raise` to explicitly generate an exception.

16. Modules vs packages?

Answer: A module is typically a Python file; a package is a collection of modules organized as a package.

17. Use of with statement?

Answer: `with` uses a context manager to handle setup and cleanup of resources safely.

18. How to connect Python to a database?

Answer: Use an appropriate driver or library such as `sqlite3`, `SQLAlchemy`, `psycopg2`, or a database-specific connector.

19. append vs extend?

Answer: `append` adds one object as a single element; `extend` adds each element from an iterable.

20. Key data engineering libraries?

Answer: pandas, NumPy, PySpark, SQLAlchemy, PyArrow and database/cloud SDKs are common.

21. Timestamp to date in pandas?

Answer: Convert with `pd.to_datetime()` and then use `.dt.date` when Python date objects are required.

22. What is duck typing?

Answer: Python focuses on whether an object supports the required operations rather than requiring a specific declared type.

23. How do you serialize objects?

Answer: Common options include pickle for Python objects and JSON for interoperable structured data.

24. What is a pandas DataFrame?

Answer: A two-dimensional labeled data structure similar to a spreadsheet or relational table.

25. Use of enumerate()?

Answer: enumerate adds a counter while iterating, commonly producing index-value pairs.

3. Apache Spark Interview Questions

For each question: answer in 1–2 sentences first. Expand only if the interviewer asks a follow-up.

1. What is Apache Spark?

Answer: A distributed data processing engine used for batch processing, streaming, SQL, machine learning, and graph workloads.

2. What are RDDs?

Answer: RDDs are immutable, partitioned, fault-tolerant distributed collections that form the lower-level Spark abstraction.

3. RDD vs DataFrame?

Answer: DataFrames provide a structured schema and allow Spark SQL optimizations such as Catalyst; RDDs provide lower-level control.

4. What is Spark SQL?

Answer: Spark SQL is Spark's module for structured data processing using SQL and the DataFrame/Dataset APIs.

5. What is lazy evaluation?

Answer: Transformations are recorded rather than immediately executed; execution begins when an action is called.

6. What are transformations?

Answer: Transformations create a new dataset from an existing one and are generally lazy, such as filter, select, and map.

7. What are actions?

Answer: Actions trigger computation and return a result or write output, such as count, collect, show, or write.

8. What is Catalyst Optimizer?

Answer: Catalyst is Spark SQL's query optimization framework that transforms logical plans into optimized execution plans.

9. What is Tungsten?

Answer: Tungsten refers to Spark execution improvements focused on efficient memory management and CPU/code generation.

10. What is DAG?

Answer: A Directed Acyclic Graph represents the sequence and dependencies of transformations used to build a Spark job.

11. Narrow vs Wide transformation?

Answer: Narrow transformations do not require data movement across partitions; wide transformations require a shuffle.

12. What is SparkSession?

Answer: SparkSession is the main entry point for Spark SQL and DataFrame operations in modern Spark.

13. What is a broadcast join?

Answer: Spark sends a small table to executors so the join can avoid a large shuffle when the broadcast fits memory.

14. What is checkpointing?

Answer: Checkpointing saves data or streaming state to reliable storage and can truncate lineage for fault tolerance.

15. What is Spark Structured Streaming?

Answer: A structured streaming engine that processes continuously arriving data using the DataFrame/Dataset API.

16. What are accumulators?

Answer: Accumulators are shared variables primarily used for aggregation/metrics from executors; workers should not use them for business logic.

17. What is partitioning?

Answer: Partitioning divides distributed data into chunks so Spark can process them in parallel.

18. How does Spark handle failures?

Answer: Spark can recompute lost partitions using lineage and rerun failed tasks.

19. cache() vs persist()?

Answer: cache() uses Spark's default storage level; persist() lets you choose a storage level.

20. What is a Spark SQL window function?

Answer: It performs calculations over a defined window of rows, such as ROW_NUMBER, RANK, LAG, and SUM OVER.

4. Hadoop Interview Questions

For each question: answer in 1–2 sentences first. Expand only if the interviewer asks a follow-up.

1. What is Hadoop?

Answer: A distributed big-data ecosystem centered around HDFS for storage and YARN for cluster resource management, with MapReduce as one processing framework.

2. Explain HDFS.

Answer: HDFS stores large files across multiple machines as blocks and replicates blocks for fault tolerance.

3. Default HDFS block size?

Answer: 128 MB is a common Hadoop 2.x+ default, but the configured value can vary by cluster/version.

4. How is HDFS fault tolerance achieved?

Answer: Through block replication across DataNodes, with replication factor commonly set to 3.

5. What is a NameNode?

Answer: The NameNode manages HDFS metadata such as file namespace, permissions, and block locations.

6. What is a DataNode?

Answer: A DataNode stores actual HDFS blocks and serves read/write requests.

7. What if NameNode fails?

Answer: Without HA the filesystem can become unavailable; HA deployments use active and standby NameNodes to reduce this risk.

8. What is Secondary NameNode?

Answer: It periodically checkpoints NameNode metadata by merging the namespace image and edit logs; it is not a standby replacement.

9. What is MapReduce?

Answer: A distributed programming model that processes data through map and reduce phases.

10. What is a Combiner?

Answer: An optional local aggregation step that can reduce data shuffled from mappers to reducers.

11. What was JobTracker?

Answer: In Hadoop 1, JobTracker handled MapReduce job scheduling and task management; Hadoop 2 moved resource management to YARN.

12. How does YARN work?

Answer: YARN separates resource management from application scheduling using components such as ResourceManager and NodeManager.

13. What is speculative execution?

Answer: It runs a duplicate copy of a slow task to reduce the impact of stragglers.

14. Hadoop 1 vs Hadoop 2?

Answer: Hadoop 2 introduced YARN and enabled Hadoop clusters to support processing frameworks beyond classic MapReduce.

15. What is rack awareness?

Answer: HDFS places replicas across racks to reduce the impact of rack or network failures.

16. How can small files be handled?

Answer: Use suitable consolidation approaches such as compaction, archive formats, or a better ingestion design to reduce metadata overhead.

17. What is a SequenceFile?

Answer: A Hadoop binary file format that stores key-value pairs and is designed for efficient MapReduce I/O.

18. Why is data locality important?

Answer: Running computation close to the data reduces network transfer and improves distributed processing efficiency.

19. How does Hadoop ensure data integrity?

Answer: HDFS uses checksums to detect corrupted data and can retrieve a valid replica.

20. What are MapReduce counters?

Answer: Counters are built-in or custom metrics used to track events and statistics during job execution.

5. Hadoop, Spark & AWS Naming Concepts

These are useful as short conversation starters. Treat the naming stories as background context rather than core technical interview topics.

1. Hadoop naming

SME-style explanation: Hadoop was named after a toy elephant belonging to Doug Cutting's son; Hadoop is not an acronym.

2. Spark naming

SME-style explanation: The name Spark reflects the idea of fast, energetic cluster computation.

3. Hive naming

SME-style explanation: Hive evokes a structured, active environment and was chosen for the data warehouse system built over Hadoop.

4. S3

SME-style explanation: S3 means Simple Storage Service and provides scalable object storage.

5. EC2

SME-style explanation: EC2 means Elastic Compute Cloud and provides resizable virtual compute capacity.

6. Lambda

SME-style explanation: Lambda is named after the Greek letter λ , commonly associated with anonymous functions and functional programming.

7. Redshift

SME-style explanation: The name references a shift toward cloud data warehousing; the branding/name also plays on the astronomical term redshift.

8. Glue

SME-style explanation: Glue is an integration service designed to connect data sources and support ETL/ELT workflows.

9. Athena

SME-style explanation: Amazon Athena is named after Athena, the Greek goddess associated with wisdom; it provides SQL querying over data such as S3 objects.

10. EMR

SME-style explanation: EMR means Elastic MapReduce and provides managed big-data frameworks including Hadoop and Spark.

11. Kinesis

SME-style explanation: Kinesis comes from Greek 'kinesis', meaning movement, fitting its role in streaming data.

12. CloudWatch

SME-style explanation: CloudWatch monitors AWS resources and applications through metrics, logs, alarms, and related observability features.

13. Step Functions

SME-style explanation: Step Functions orchestrates workflows as a sequence of states and steps, often coordinating serverless services.

6. High-Value Follow-Up Questions

SQL: When would you use a window function instead of GROUP BY?

Answer: Use GROUP BY when you want one row per group; use a window function when you need aggregate/ranking information while retaining individual rows.

SQL: Why can an index improve SELECT performance but hurt writes?

Answer: The index reduces lookup work for reads, but inserts/updates/deletes must also maintain the index structure.

SQL: Why is UNION ALL often faster than UNION?

Answer: UNION ALL does not perform duplicate elimination.

Python: Why use generators for large files?

Answer: They produce values lazily, avoiding loading the entire dataset into memory.

Python: When would you choose a tuple?

Answer: When the collection should not be modified and immutability is useful.

Spark: Why does a shuffle matter?

Answer: A shuffle moves data between executors/partitions and can become a major performance bottleneck.

Spark: Why avoid collect() on a large DataFrame?

Answer: collect() brings all rows to the driver and can cause driver memory failure.

Spark: When would you broadcast a table?

Answer: When one side of a join is sufficiently small to fit safely in executor memory.

Spark: cache() vs persist()?

Answer: cache uses the default persistence level; persist allows selecting a specific storage level.

Hadoop: NameNode vs DataNode?

Answer: NameNode manages HDFS metadata; DataNodes store the actual blocks.

Hadoop: Why is replication needed?

Answer: It allows data to remain available when a DataNode or other infrastructure component fails.

AWS: What is the difference between S3 and EBS?

Answer: S3 is object storage; EBS provides block storage volumes attached to compute instances such as EC2.

7. 10-Minute Final Revision Checklist

- ☐ SQL: joins, GROUP BY/HAVING, subqueries, CTEs, window functions, RANK vs DENSE_RANK, indexes, NULL/COALESCE, UNION, transactions.
- ☐ Python: list/tuple, dict, functions, *args/**kwargs, lambda, generators, exceptions, copy/deepcopy, pandas basics.
- ☐ Spark: DataFrame vs RDD, lazy evaluation, transformations/actions, DAG, shuffle, partitioning, broadcast join, cache/persist, SparkSession.
- ☐ Hadoop: HDFS, NameNode/DataNode, replication, blocks, YARN, MapReduce, rack awareness, data locality.
- ☐ AWS: S3, EC2, Lambda, Glue, Athena, EMR, Kinesis, CloudWatch, Step Functions.
- ☐ Interview delivery: answer directly first; use a practical example; don't over-explain unless asked.

Important accuracy notes

- SQL behavior can vary by database. In particular, rollback behavior of TRUNCATE is database-specific.

- Normal SQL views are virtual definitions; indexed/materialized view support depends on the database.
- Spark's Structured Streaming is the modern structured streaming API; DStreams are the older Spark Streaming API.
- HDFS block size and replication factor are configuration defaults, not universal constants.